

Forecasting Approaches for Limited Data

Introduction

Scenario:

- You have 5 observations over ~10-15 years
- Measurements are irregular (not monthly, not annual)
- You need a forecast 3-5 years ahead

Why this is hard:

- Not enough data to reliably estimate many parameters
- No repeating seasonality cycles to learn from
- High sensitivity: one point can change the whole forecast

Many real-world outcomes look like this

- Survey-based indicators
- Expensive measurements
- Emerging markets / new products / rare events

What “Limited Data” Really Means

Limited data can mean:

- **Small T**: very few time points
- **Irregular spacing**: gaps of different lengths
- **Measurement noise**: survey error / revisions
- **Bounded targets**: rates in $[0,1]$ or $[0,100]$
- **Regime changes**: policy shifts, shocks, interventions

Key mindset: we compensate with structure + assumptions + transparency

The Approaches

Three practical approaches for sparse data:

- Trend Regression - Fit a simple trend line
- Scenario-Based Forecasting - Define multiple futures
- External Drivers - Use related variables with more data

Plus: Visualizing Uncertainty - Be honest about what you don't know

Approach 1 - Trend Regression

Idea: Fit the simplest model that captures the pattern

Safe options (with 5 points):

- Linear trend in time: $y = a + b \cdot t$
- Piecewise trend (only if you can justify a break)
- Robust regression (down-weight outliers)

Why it works here:

- Few parameters
- Clear assumptions
- Easy to explain and stress-test

Bounded Outcomes - Don't Let Forecasts Go Below 0 or Above 100

If your target is a rate/percent, enforce realism:

Option A (recommended): Logit transform

- Convert percent to proportion $p \in (0,1)$
- Model: $\text{logit}(p) = a + b \cdot t$
- Invert back to percent $\rightarrow$ forecasts stay within bounds

Option B: Logistic growth curve

- Useful if saturation is expected (approaches a ceiling)

Why this matters:

- A plain linear trend can overshoot 100% quickly

Irregular Time Gaps - How to Handle Them

Regression doesn't require regular spacing.

Do this:

- Define $t = \text{year} - \text{year}_0$ (e.g., $2008 \rightarrow t=0$)
- Fit using t as the predictor

Avoid:

- Methods that assume repeated periodic patterns (seasonality) when none exists

Approach 2 - Scenario-Based Forecasting

Idea: Instead of one forecast, define multiple plausible futures

Typical scenarios:

- **Optimistic:** Favorable conditions, interventions succeed
- **Base case:** Current trends continue
- **Pessimistic:** Challenges arise, slowdowns occur

How to define scenarios:

- Adjust the trend slope
- Adjust the starting point
- Incorporate known future events

Defining Scenarios

Scenarios should be **plausible**, not arbitrary.

How to justify them:

- Use historical ranges from comparable contexts
- Use domain constraints (e.g., saturation)
- Use known future plans/risks (policy, funding, shocks)

Example parameterization:

- Base: fitted slope
- Optimistic: slope +20% (or faster convergence to ceiling)
- Pessimistic: slope -30% (or earlier saturation / stall)

Approach 3 - Using External Drivers

Idea: Borrow information from related variables that have more data

Example:

- School completion rate: 5 data points
- GDP per capita: annual data (15 points)
- Adult literacy rate: annual data (15 points)

If completion rate correlates with GDP:

- Use GDP as a predictor
- More data → more stable estimates

Choosing External Drivers

Good drivers:

- Logically related to your target
- Available at higher frequency
- Lead (not lag) your target variable

Caution:

- Correlation $\neq$ causation
- Don't overfit with too many drivers
- Document your assumptions

Visualizing Uncertainty

With 5 data points, uncertainty is unavoidable.

Type 1: Statistical uncertainty

- “Given this model, how uncertain is the forecast?”

Type 2: Structural uncertainty

- “What if this model is wrong?”
- “What if the future regime changes?”

Tools:

- Confidence / prediction intervals (Type 1)
- Scenarios (Type 2)

Confidence vs Prediction Intervals

Confidence interval (CI):

- Uncertainty about the mean trend

Prediction interval (PI):

- Uncertainty about future observed values
- Wider than CI (and usually what stakeholders actually need)

Practical approach:

- Use `statsmodels.get_prediction()`
- Plot shaded PI band + scenario lines

Key Takeaways

- Limited data $\neq$ no forecast - simpler methods work
- Trend regression is your friend with sparse data
- For rates, use bounded models (logit/logistic)
- Scenarios communicate range of possibilities
- External drivers can strengthen forecasts
- Uncertainty is honest - wide bands are appropriate
- Document everything - assumptions matter

A practical workflow:

1. **Explore:** plot points; check bounds; note gaps
2. **Fit trend:** linear or bounded (logit/logistic)
3. **Add drivers:** only if defensible and available
4. **Define scenarios:** optimistic/base/pessimistic
5. **Quantify uncertainty:** prediction intervals + scenario range
6. **Visualize clearly:** points + line + bands + scenarios
7. **Document assumptions:** what must be true for forecast to hold

What NOT to Do

- Fit high-parameter models to 5 points
- Report a single point forecast with no uncertainty
- Add many predictors (guaranteed overfit)
- Extrapolate far beyond the data without scenario framing
- Hide assumptions or ignore bounds

Any questions?